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12.6% Fewer Vehicles, Zero Regressions: An Audit-First ALNS for Electric-Vehicle Routing

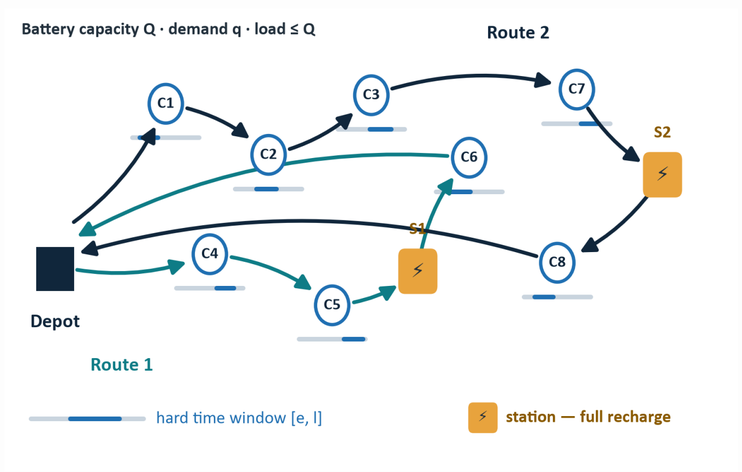
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Motivation

E-VRPTW: battery-limited EVs, hard time windows

Fleet size first in the lexicographic objective

Feasibility hinges on exact charging decisions



Objective — reproduce SSG (2014), then cut fleet size with zero regressions

Benchmark — SSG (2014) · 12 instances · 5–100 customers · C / R / RC

Model — battery-limited EVs · en-route charging · hard time windows

Toolchain — Python · numba · CPLEX · Gurobi · HiGHS · OR-Tools (lockfile-pinned)

Method Overview

INITIAL SOLUTION
greedy insertion · one route per seed customer cluster

DESTROY
constraint-guided removal:
station pressure · TW conflict · energy detour

REPAIR
vehicle-count-aware greedy insertion
(existing routes first, new route last)

PREFILTER
capacity · optimistic time · optimistic energy
(cheap, shared, screens before exact work)

EXACT CHARGING
labeling subproblem,
solved per changed route only

ACCEPT?
simulated annealing —
never accepts more vehicles

iterate × 1000 (30 s budget)

lexicographic objective: vehicles ≫ distance ≫ charging time

Constraint-guided ALNS engine — vehicle-first search with exact charging subproblems

Exact charging dominates cost: 1,687–2,161 calls per 30 s run

Acceleration shipped: incremental cache · exact-deadline · parallel control

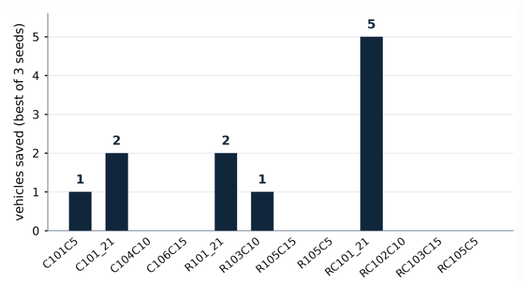
KEY MECHANISMS

- Vehicle-count-aware repair: eliminate routes first; fleet can fall, never rise
- Optimistic prefilters (capacity · time · energy) screen before exact evaluation
- Constraint-guided destroy: station pressure, TW conflicts, energy detours
- Exact charging subproblems re-solve only the routes that changed

AUDIT PROTOCOL

- Lexicographic objective — vehicles first; acceptance rejects any fleet-size increase
- Independent review CLI replays solutions, event logs & hashes before publication

VEHICLE SAVINGS



Best of 3 seeds per instance — total fleet 87 → 76 vehicles (−12.6%)

Experimental Settings

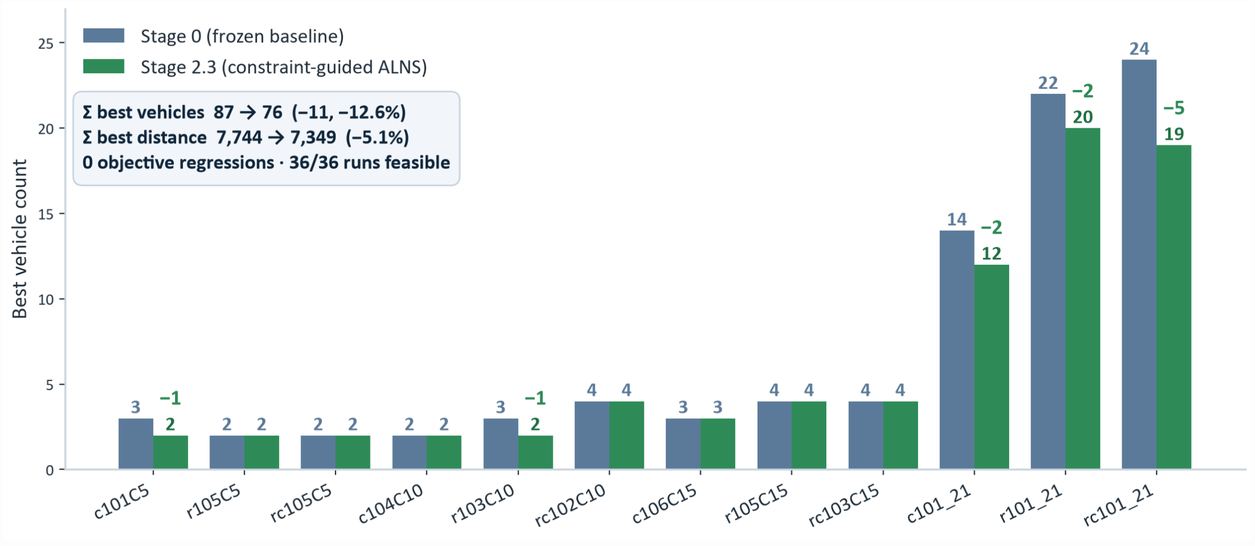
every stage gated by an independent CLI review that replays raw JSON + event logs

- **Stage 0** — Baseline reproduction of SSG (2014) — 36/36 feasible, frozen forever — READY
- **Stage 1** — Lexicographic objective: vehicles ≫ distance ≫ charging time — READY
- **Stage 2.1–2.3** — Route reduction → constraint-guided ALNS; −12.6% vehicles · −5.1% distance · zero regressions — READY
- **Stage 3.0–3.4** — Cheap screening · cache · exact deadline · control parallel; −22% median exact-charging CPU (C set) — READY
- **Stage 4** — Adaptive operator weights — wins on 7/36 runs (144/144 re-verified) — READY
- **Stage 5.1** — Honest BKS audit — 92/124 entries model-incompatible, no fake gaps — READY

Instances	12
Seeds	3
Time limit	30 s
Iterations	1,000
Thread	1

Stage 0 frozen & hash-verified — every number replays from raw evidence

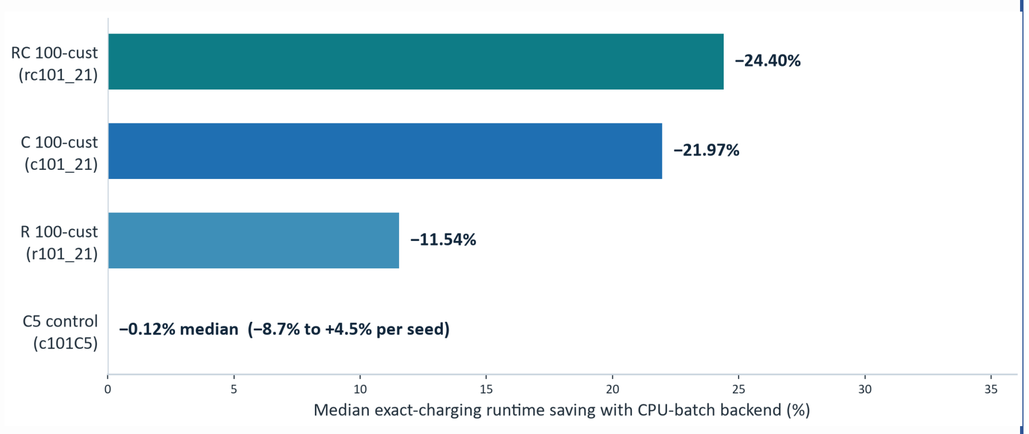
Key Results



Vehicle-first search cuts fleet on 5/12 instances — 87 → 76 vehicles, −5.1% distance, 36/36 feasible

Per-family best — C 22→19 · R 31→28 · RC 34→29 · mean fleet 90.7 → 77.0

Key Quantitative Highlights



−22.0% exact-charging CPU (median, C instances) — CPU-batch default after audit

Batch pilot covers 4 instances (speedups 1.00–1.32×) — full-scope acceleration remains a Stage 3 target

Why It Works & Honest Audit

✓WHY IT WORKS

- ✓Vehicle-count-aware repair — fleet can fall, never rise
- ✓Constraint-guided destroy targets station pressure · TW conflicts · energy detours
- ✓Dynamic removal size escalates on stagnation (17,892 logged)
- ✓Stage-gated: two independent complete reruns must agree

▲HONEST BOUNDARIES

- ▲92/124 published BKS entries are model-incompatible — no gap claims against them
- ▲C5 control: −0.12% median (−8.7%…+4.5%) — speedup is structure-dependent, not universal
- ▲9/36 runs overran the 30 s budget by ≤ 0.02 s — all kept, with events
- ▲No unattested baselines — every number replays from raw solution JSON + event logs

EVIDENCE AT SCALE

12,121	200,731	20	0
evidence files on disk	operator event rows replayed	real failure cases kept	synthetic cases in audit

[1] Schneider, Stenger & Goeke (2014) Transp. Sci. 48(4):500–520 [2] Pisinger & Ropke (2006) Transp. Res. B
[3] Keskin & Çatay (2016) Transp. Res. E [4] Hottung, Kwon & Tierney (2022) EJOR

Scan for code & evidence

−5.1% total distance | **36/36 feasible formal runs** | **−22.0% exact-charging CPU**

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